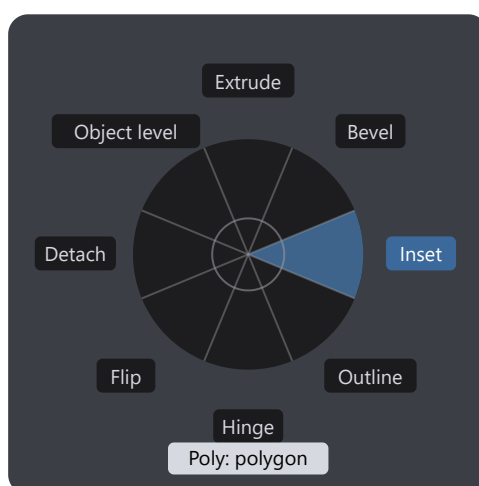


MarkingForge

Reference - every option, every command, tips and troubleshooting

3DS MAX 2027 · VERSION 1.0.8



REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears only when you hold the key and wait, to remind you where things are. After a few days your hand knows the directions and you stop looking.

Installing is covered by the **Installation Guide**. This document explains how everything works once it is installed: the gestures, the 24 built-in dials, every control of the menu editor, every experimental feature, the configuration file, the colours, the MAXScript commands, tips and tricks, and what to do when something does not behave.

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1. Concepts

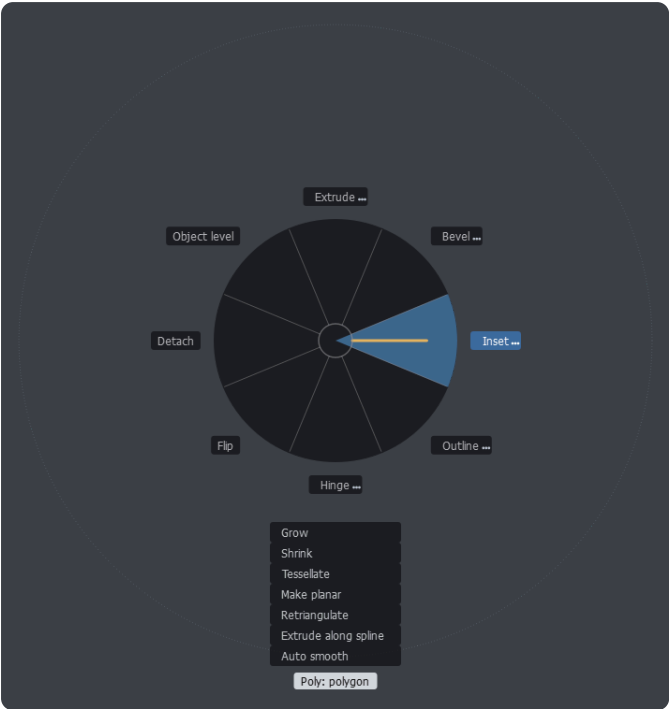
Term	Meaning
Dial	The round menu that appears around the cursor while you hold a key. Eight directions, a centre that cancels, optionally a list of rows under it.
Slot	One of the 24 dials. Slot N is the action <i>MarkingForge: Menu N</i> in the Hotkey Editor; its key or mouse button opens it.
Direction	N, NE, E, SE, S, SW, W, NW. Each holds one item - or nothing (an empty direction cancels).
Item	What a direction runs: a 3ds Max action, a macroscript, your own MAXScript, a submenu or a value slider (chapter 8).
List under the dial	Up to twelve rows drawn below the ring for things that do not fit in eight directions. Rows are clicked, not aimed at.
Variant	Alternative contents of the same dial with a condition - the object class, the sub-object level, the window under the cursor... The first variant whose condition matches wins; <i>Base</i> is shown when none does (chapter 6).
Content source	Instead of fixed items, a dial can show live data: the modifier stack, the undo list, recent commands, selection sets (chapter 7).
Hotbox	A slot laid out as tiles of 3ds Max's main menu instead of eight directions.
Default menu	What a slot without its own menu shows. Every slot of the built-in layout has its own menu, so the default only matters for slots you empty yourself.
Caddy	The small settings window 3ds Max shows for Chamfer, Extrude, Inset and similar commands. The dial can open it or skip it (chapter 2).

2. Gestures - everything a dial reacts to

Gesture	What happens	Notes
Hold the key	The dial appears around the cursor.	The centre stays where it opened - the dial does not follow the mouse.
Move towards a direction, release the key	That direction runs.	The wedge and its caption light up while you aim, so you see the choice before committing.
Press and move at once (flick)	The direction runs without the dial being drawn.	The fastest way. Works as soon as your hand knows the direction. The context - the object and the window under the cursor - is read where the gesture began, so a flick does exactly what the held dial shows.
Tap (press and release without moving)	Nothing, unless the dial has a tap item.	Set in the editor: <i>A tap runs</i> . Uses the context variant that matches, like the dial.
Release in the centre	Cancel - nothing runs.	The small circle in the middle is the dead zone.
Release over an empty direction	Cancel.	An empty direction is drawn with a dashed outline and never lights up.
Esc, or the right mouse button	Cancel and close.	
Move far past the ring (the settings rim)	Reverses the settings choice: an item that normally runs at once opens its caddy, and the other way round.	The rim is the dotted circle outside the dial; it brightens once the cursor is past it. The dots at an item's edge show which way the release will go. Coming back inside before releasing undoes the choice.
Move through a submenu direction	The submenu opens as a new dial where you are.	Up to three levels. A path trail shows how you got there.
Click a row of the list under the dial	That row runs.	Hold the key, move down to the list, point, click.
Cross the ring and come back to the dashed circle	The item is queued (a chain). Releasing in the dashed circle runs the whole queue as ONE undo; releasing at the very centre or over an empty direction discards it.	Experimental <i>multiPick</i> - off by default. The dashed circle round the centre marks the return zone and turns green while a queue waits; the dot in the middle always cancels. The queue shows as a numbered row above the dial.
Tap with no tap item on the dial	Runs the suggested next command, when pair learning knows one and it sits on this dial.	Experimental <i>pairLearning</i> - on by default. The suggestion is the outlined tile; an explicit tap item always wins.
Keep moving through a value direction	The distance sets a number; the release commits it.	Experimental <i>dragValue</i> - off by default.
Start typing while the dial is open	The dial becomes a search over all 3ds Max actions. It pins itself, so you can let go of the key; Enter runs, Esc closes.	Experimental <i>gestureSearch</i> - off by default.

The dial never takes keyboard focus from 3ds Max. That is why it works over the Slate editor and Track View without changing what they do - and why switching to another program closes it as a cancel.

3. Reading the dial



A dial aimed east.

You see	It means
Dark disc with eight wedges	The eight directions.
Lit wedge and tile	What a release will run now.
Dashed outline	An empty direction - releasing there cancels.
Red caption	A broken item: the action or macro does not exist in this 3ds Max (for example a plugin that is not installed). Releasing does nothing.
Grey caption	Unavailable right now (the command is disabled in this context).
Dot on a tile	A toggle that is on (checked).
> on a tile	A submenu.
... at a tile's edge	The command has a settings dialog (caddy).
Outlined tile	The default direction - what a tap runs (with pair learning: the command you usually run next), and Enter on a pinned dial.
Dotted circle outside the dial	The settings rim - brighter once you are past it (chapter 2).
Dashed circle round the centre	A chain is being built - come back inside it to queue the item.
Footer	The dial's name and, when it matters, why it is empty.

4. The 24 built-in dials

The layout a fresh installation starts with (**Load default dials...** in the editor brings it back). The grids show directions as they sit on screen, the dial's name in the centre. Opposite directions hold opposite operations - All / None, Grow / Shrink, Hide / Unhide - so the hand learns one movement and its reverse.

The poly layout, level by level

The same on an Editable Poly and under an Edit Poly modifier. NW always goes back to object level.

Level	N	NE	E	SE	S	SW	W	NW
Object	Vertex	Edge	Border	Polygon	Element	Attach	+ TurboSmooth	Collapse stack
Vertex	Chamfer	Weld	Connect	Break	Extrude	Remove	Target Weld	Object level
Edge	Connect	Chamfer	Loop	Ring	Extrude	Weld	Remove	Object level
Border	Cap	Chamfer	Extrude	Connect	Create shape	Shrink	Grow	Object level
Polygon	Extrude	Bevel	Inset	Outline	Hinge	Flip	Detach	Object level
Element	Detach	Hide	Flip	Unhide all	Hide unselected	Shrink	Grow	Object level

Alt+1 Modelling

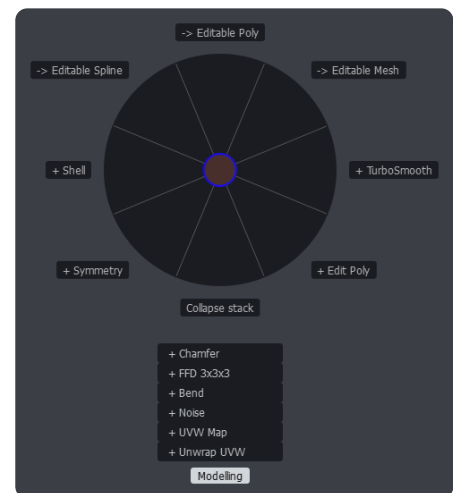
NW -> Editable Spline	N -> Editable Poly	NE -> Editable Mesh
W + Shell	Modelling	E + TurboSmooth
SW + Symmetry	S Collapse stack	SE + Edit Poly

List under the dial: + Chamfer · + FFD 3x3x3 · + Bend · + Noise · + UVW Map · + Unwrap UVW

Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor
- **Track View** - when cursor over Track View
- **Particle View** - when cursor over Particle View
- **Scene Explorer** - when cursor over the Scene Explorer
- **Modifier stack** - when cursor over the command panel
- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel
- **Edit Poly: vertex** - when Edit Poly is current in the Modify panel, sub-object level: Vertex
- **Edit Poly: edge** - when Edit Poly is current in the Modify panel, sub-object level: Edge
- **Edit Poly: border** - when Edit Poly is current in the Modify panel, sub-object level: Border / Segment
- **Edit Poly: polygon** - when Edit Poly is current in the Modify panel, sub-object level: Polygon
- **Edit Poly: element** - when Edit Poly is current in the Modify panel, sub-object level: Element
- **Edit Poly: object** - when Edit Poly is current in the Modify panel
- **Poly: vertex** - when the object is an Editable Poly, sub-object level: Vertex
- **Poly: edge** - when the object is an Editable Poly, sub-object level: Edge
- **Poly: border** - when the object is an Editable Poly, sub-object level: Border / Segment
- **Poly: polygon** - when the object is an Editable Poly, sub-object level: Polygon
- **Poly: element** - when the object is an Editable Poly, sub-object level: Element
- **Poly: object** - when the object is an Editable Poly
- **Spline: vertex** - when a shape (spline) is selected, sub-object level: Vertex
- **Spline: spline** - when a shape (spline) is selected, sub-object level: Spline
- **Spline** - when a shape (spline) is selected
- **Nothing selected** - when nothing is selected

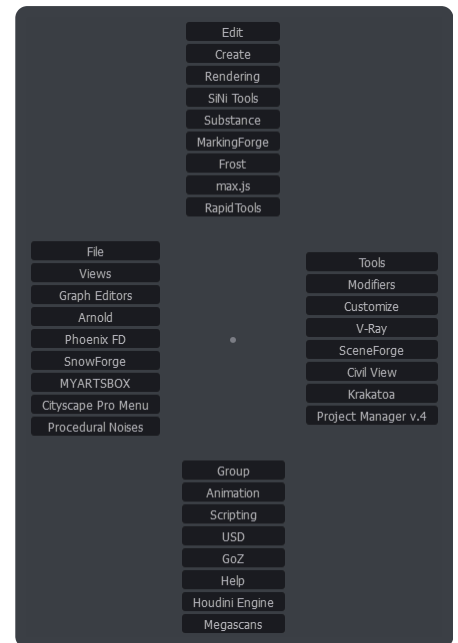
dial 1



Alt+2 Hotbox - Max's menu

Shows tiles built from Max's whole main menu - including menus other plugins add. Point at the entry and release, or click it; your selection is not changed.

dial 2



Alt+3 Create

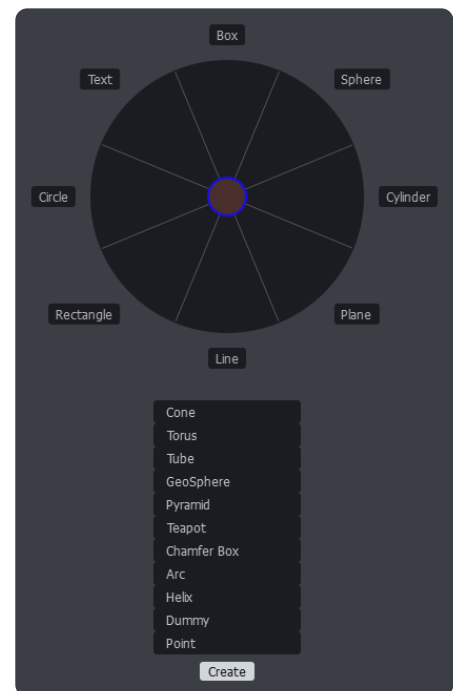
NW Text	N Box	NE Sphere
W Circle	Create	E Cylinder
SW Rectangle	S Line	SE Plane

List under the dial: Cone · Torus · Tube · GeoSphere · Pyramid · Teapot · Chamfer Box · Arc · Helix · Dummy · Point

Changes with the situation (first match wins):

- **Create shapes** - when a shape (spline) is selected

dial 3



Alt+4 View

NW Zoom extents	N Zoom extents all	NE Zoom all selected
W Wireframe / shaded	View	E Maximize viewport
SW Zoom selected	S Isolate	SE Edged faces

List under the dial: Command panel · Main toolbar · Expert mode · Statistics · Safe frames · Cross-hair cursor · Solid background · Undo view change · Redo view change

dial 4



Alt+5 Selection

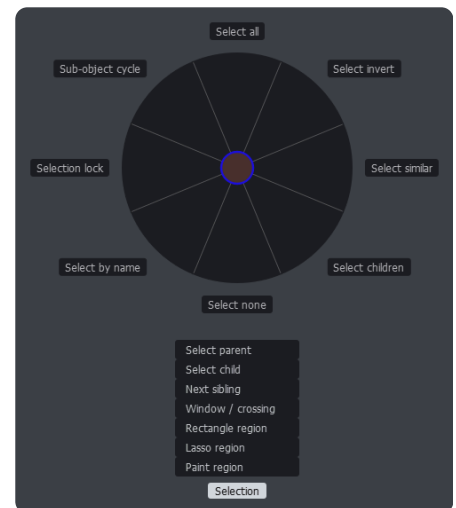
NW Sub-object cycle	N Select all	NE Select invert
W Selection lock	Selection	E Select similar
SW Select by name	S Select none	SE Select children

List under the dial: Select parent · Select child · Next sibling · Window / crossing · Rectangle region · Lasso region · Paint region

Changes with the situation (first match wins):

- **Selection** - when Edit Poly is current in the Modify panel, sub-object level: object level
- **Edit Poly: selection** - when Edit Poly is current in the Modify panel
- **Selection** - when the object is an Editable Poly, sub-object level: object level
- **Poly: selection** - when the object is an Editable Poly

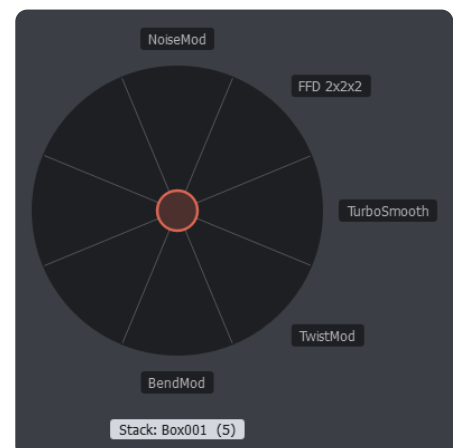
dial 5



Alt+6 Modifier stack

Filled live from the modifier stack of the selected object (top eight).

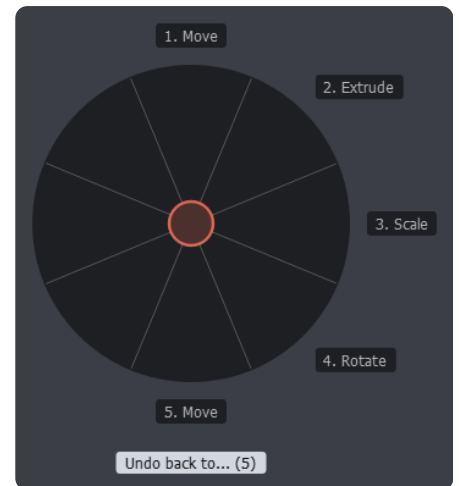
dial 6



Alt+7 Undo

Filled live from Max's undo list - N undoes one step, NE two, and so on.

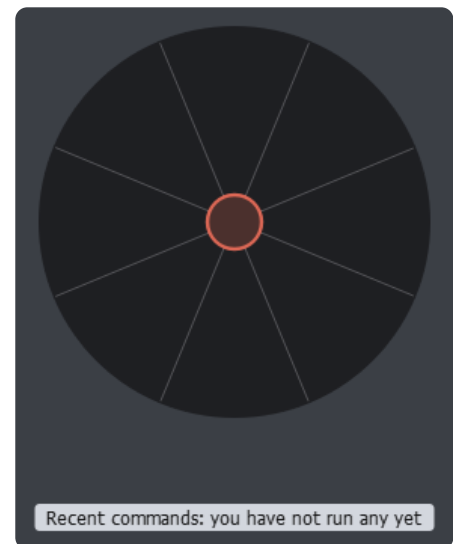
dial 7



Alt+8 Recent commands

Filled live from the eight commands you ran most recently.

dial 8

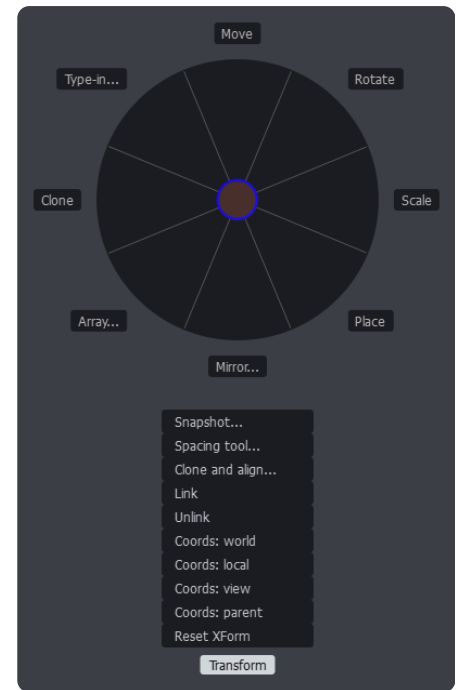


Ctrl+Alt+1 Transform

NW Type-in...	N Move	NE Rotate
W Clone	Transform	E Scale
SW Array...	S Mirror...	SE Place

List under the dial: Snapshot... · Spacing tool... · Clone and align... · Link · Unlink · Coords: world · Coords: local · Coords: view · Coords: parent · Reset XForm

dial 9



Ctrl+Alt+2 Modifiers

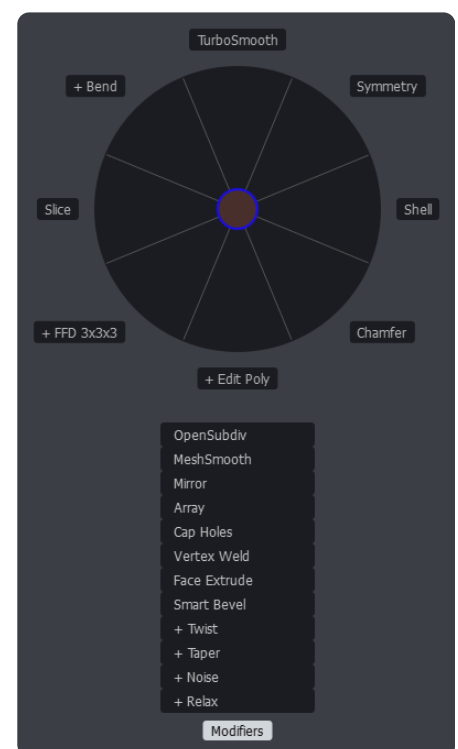
NW + Bend	N TurboSmooth	NE Symmetry
W Slice	Modifiers	E Shell
SW + FFD 3x3x3	S + Edit Poly	SE Chamfer

List under the dial: OpenSubdiv · MeshSmooth · Mirror · Array · Cap Holes · Vertex Weld · Face Extrude · Smart Bevel · + Twist · + Taper · + Noise · + Relax

Changes with the situation (first match wins):

- **Shape modifiers** - when a shape (spline) is selected

dial 10



Ctrl+Alt+3 Show and hide

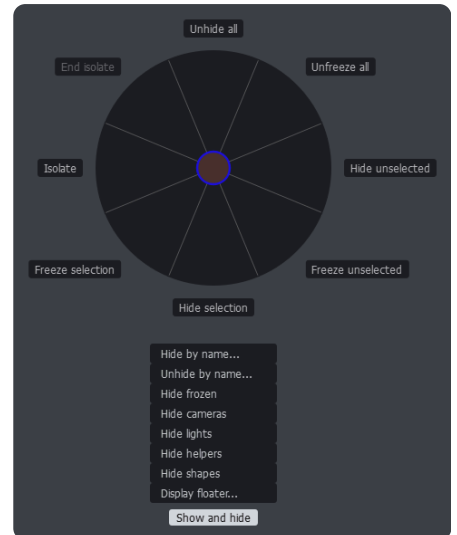
dial 11

NW End isolate	N Unhide all	NE Unfreeze all
W Isolate	Show and hide	E Hide unselected
SW Freeze selection	S Hide selection	SE Freeze unselected

List under the dial: Hide by name... · Unhide by name... · Hide frozen · Hide cameras · Hide lights · Hide helpers · Hide shapes · Display floater...

Changes with the situation (first match wins):

- **Show and hide** - when the object is an Editable Poly, sub-object level: object level
- **Poly: show and hide** - when the object is an Editable Poly

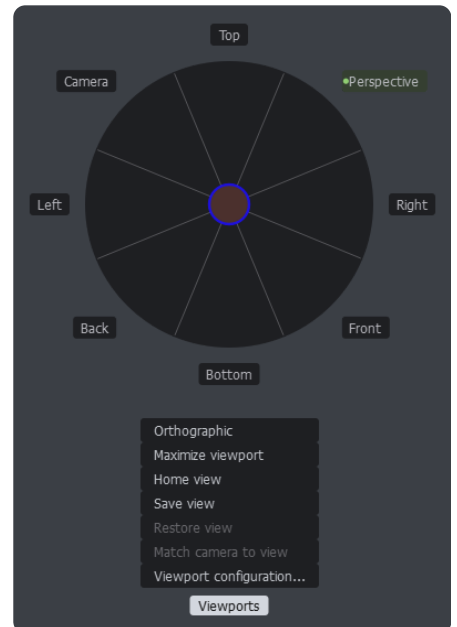


Ctrl+Alt+4 Viewports

dial 12

NW Camera	N Top	NE Perspective
W Left	Viewports	E Right
SW Back	S Bottom	SE Front

List under the dial: Orthographic · Maximize viewport · Home view · Save view · Restore view · Match camera to view · Viewport configuration...

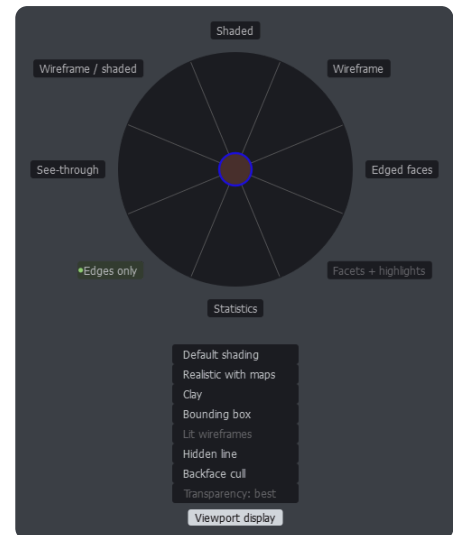


Ctrl+Alt+5 Viewport display

dial 13

NW Wireframe / shaded	N Shaded	NE Wireframe
W See-through	Viewport display	E Edged faces
SW Edges only	S Statistics	SE Facets + highlights

List under the dial: Default shading · Realistic with maps · Clay · Bounding box · Lit wireframes · Hidden line · Backface cull · Transparency: best

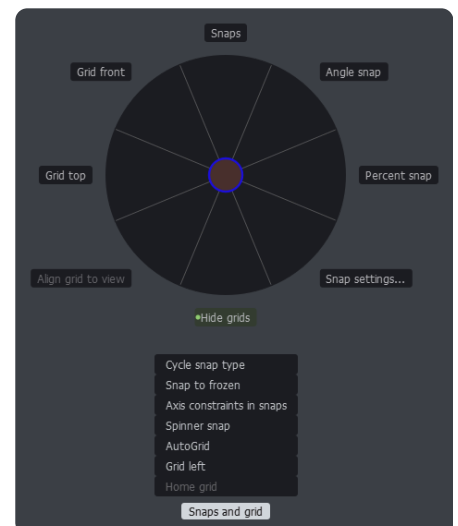


Ctrl+Alt+6 Snaps and grid

dial 14

NW Grid front	N Snaps	NE Angle snap
W Grid top	Snaps and grid	E Percent snap
SW Align grid to view	S Hide grids	SE Snap settings...

List under the dial: Cycle snap type · Snap to frozen · Axis constraints in snaps · Spinner snap · AutoGrid · Grid left · Home grid



Ctrl+Alt+7 Align and pivot

dial 15

NW Working pivot to view	N Quick align	NE Normal align...
W Pin working pivot	Align and pivot	E Align to view...
SW Place pivot surface	S Edit working pivot	SE Align camera

List under the dial: Align... · Place pivot view · Use working pivot · Reset working pivot · Center pivot · Pivot to origin · Reset XForm

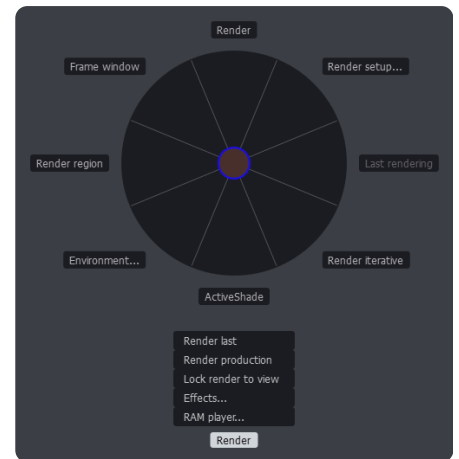


Ctrl+Alt+8 Render

dial 16

NW Frame window	N Render	NE Render setup...
W Render region	Render	E Last rendering
SW Environment...	S ActiveShade	SE Render iterative

List under the dial: Render last · Render production · Lock render to view · Effects... · RAM player...

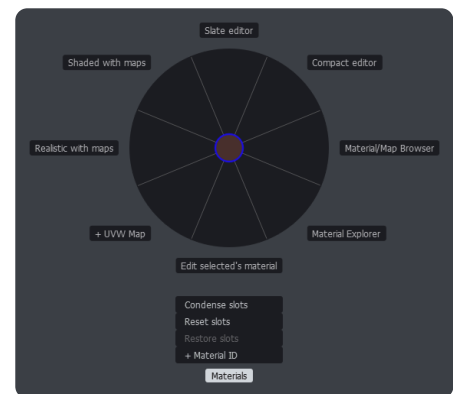
**Shift+Alt+1 Materials**

dial 17

NW Shaded with maps	N Slate editor	NE Compact editor
W Realistic with maps	Materials	E Material/Map Browser
SW + UVW Map	S Edit selected's material	SE Material Explorer

List under the dial: Condense slots · Reset slots · Restore slots · + Material ID
Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor

**Shift+Alt+2 Animation**

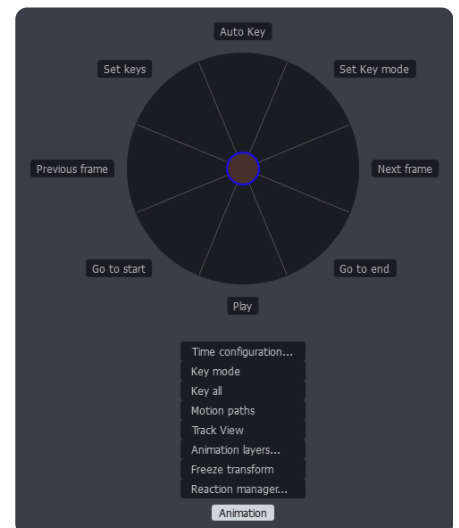
dial 18

NW Set keys	N Auto Key	NE Set Key mode
W Previous frame	Animation	E Next frame
SW Go to start	S Play	SE Go to end

List under the dial: Time configuration... · Key mode · Key all · Motion paths · Track View · Animation layers... · Freeze transform · Reaction manager...

Changes with the situation (first match wins):

- **Track View** - when cursor over Track View



Shift+Alt+3 Scene

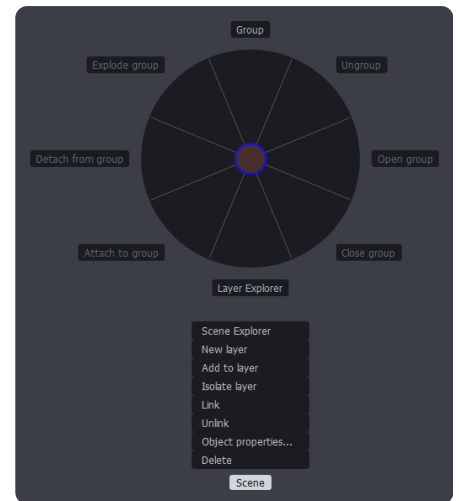
dial 19

NW Explode group	N Group	NE Ungroup
W Detach from group	Scene	E Open group
SW Attach to group	S Layer Explorer	SE Close group

List under the dial: Scene Explorer · New layer · Add to layer · Isolate layer · Link · Unlink · Object properties... · Delete

Changes with the situation (first match wins):

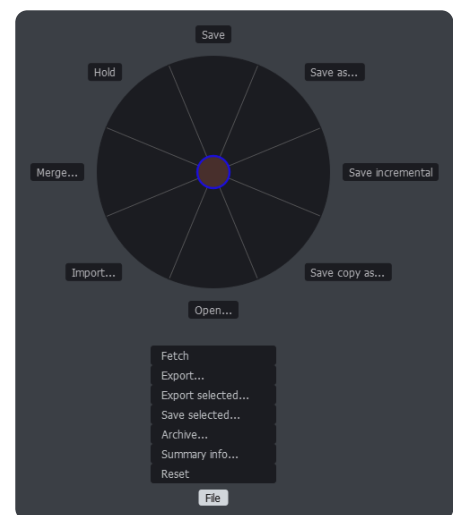
- **Scene Explorer** - when cursor over the Scene Explorer

**Shift+Alt+4 File**

dial 20

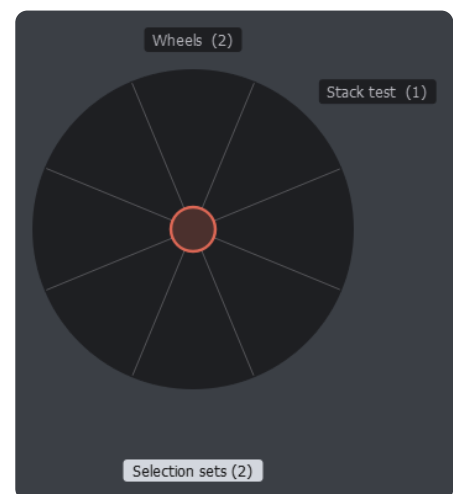
NW Hold	N Save	NE Save as...
W Merge...	File	E Save incremental
SW Import...	S Open...	SE Save copy as...

List under the dial: Fetch · Export... · Export selected... · Save selected... · Archive... · Summary info... · Reset

**Shift+Alt+5 Selection sets**

dial 21

Filled live from the named selection sets of the scene.



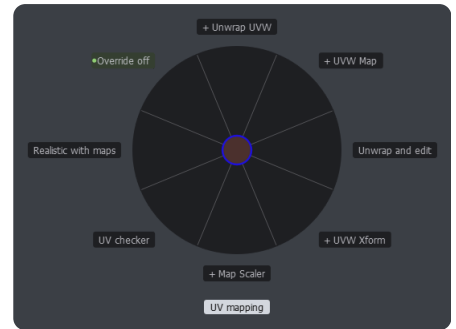
Shift+Alt+6 UV mapping

dial 22

NW Override off	N + Unwrap UVW	NE + UVW Map
W Realistic with maps	UV mapping	E Unwrap and edit
SW UV checker	S + Map Scaler	SE + UVW Xform

Changes with the situation (first match wins):

- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel

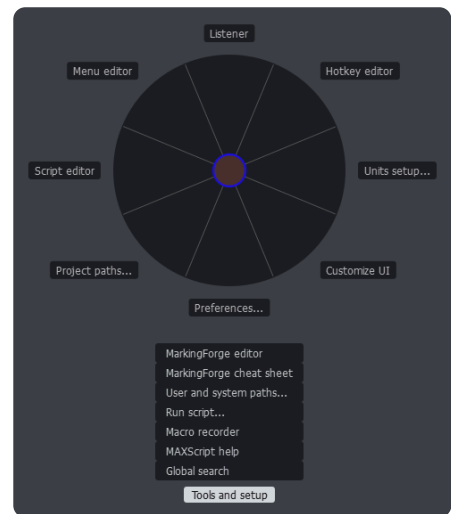


Shift+Alt+7 Tools and setup

dial 23

NW Menu editor	N Listener	NE Hotkey editor
W Script editor	Tools and setup	E Units setup...
SW Project paths...	S Preferences...	SE Customize UI

List under the dial: MarkingForge editor · MarkingForge cheat sheet · User and system paths... · Run script... · Macro recorder · MAXScript help · Global search



Shift+Alt+8 Lights and cameras

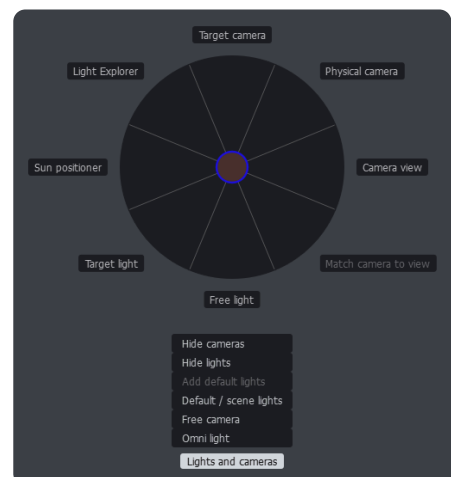
dial 24

NW Light Explorer	N Target camera	NE Physical camera
W Sun positioner	Lights and cameras	E Camera view
SW Target light	S Free light	SE Match camera to view

List under the dial: Hide cameras · Hide lights · Add default lights · Default / scene lights · Free camera · Omni light

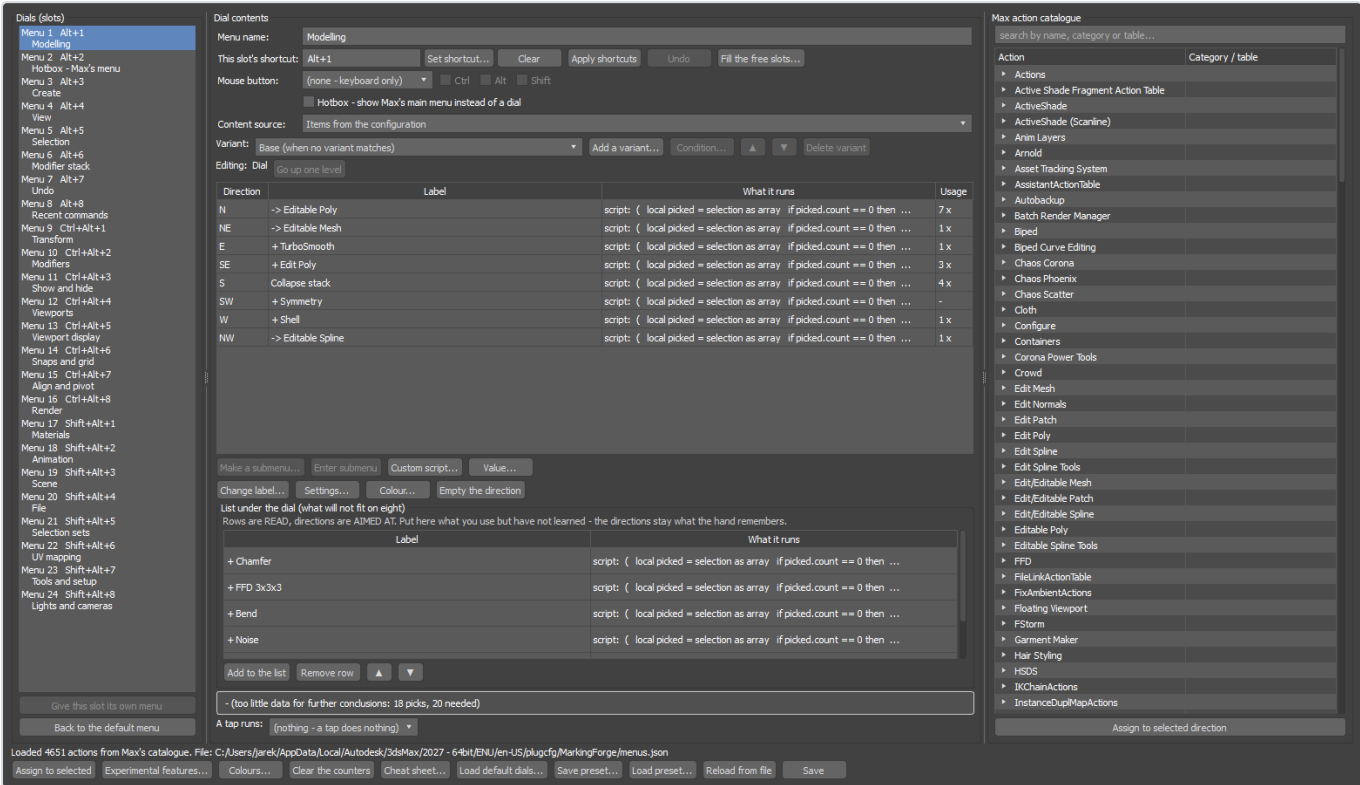
Changes with the situation (first match wins):

- **Camera** - when a camera is selected
- **Light** - when a light is selected



5. The menu editor, control by control

MarkingForge > Menu editor... Changes are kept in the editor until you press **Save**; the plugin then reloads within about 150 ms, without restarting 3ds Max.



The menu editor.

Left - the dials

Control	What it does
Dials (slots)	The 24 slots with their shortcut and menu name. Pick one to edit it.
Give this slot its own menu	Makes an independent copy of the default menu for this slot. Needed before editing a slot that shows the default.
Back to the default menu	Deletes this slot's own menu - the slot shows the default again. What it held is lost.

Middle - the dial

Control	What it does
Menu name	The name shown in the dial's footer, in the cheat sheet and next to the slot.
This slot's shortcut	Shows the key. Set shortcut... opens a window: press the keys there (Ctrl, Alt, Shift in any combination) and click Set . While it is open the dials and 3ds Max's own shortcuts are paused, so a key that already opens a dial can be recorded; it then moves from that dial to this one. The window also says when 3ds Max uses the key for another command. Clear (or Backspace) removes the key. Nothing reaches 3ds Max until Apply shortcuts .
Apply shortcuts	Writes the shortcuts of all 24 slots into 3ds Max at once. Shortcuts of other commands are kept. A backup is made first.
Undo (next to it)	Restores the shortcuts from before the last Apply.

Fill the free slots...	Offers Alt+1..8 / Ctrl+Alt+1..8 / Shift+Alt+1..8 for every slot without a shortcut and lists keys already used by something else before writing.
Mouse button (+ Ctrl / Alt / Shift)	Opens the dial by holding a mouse button with a modifier. A modifier is required (chapter 9). A button opens ONE dial, so on a slot that still shows the default menu the editor first offers to give the slot its own menu.
Hotbox	Turns the slot into a hotbox of 3ds Max's main menu; the directions are then not used.
Content source	Items from the configuration, or a live source (chapter 7).
Variant	Choose which contents you edit: <i>Base</i> or one of the variants.
Add a variant...	New contents with a condition - starts as a copy of what you see.
Condition...	Edits the selected variant's condition (chapter 6).
▲ / ▼	Moves the variant up or down. The first match wins - keep specific conditions above general ones.
Delete variant	Removes the variant and its contents; the base stays.
Go up one level	Leaves a submenu you are editing.
Directions table	N ... NW with label, what runs and the usage count. Fill a row by dragging a command from the catalogue, double-clicking a command, or Assign to selected direction . Double-click a submenu row to step inside.
Make a submenu...	Turns the selected direction into a submenu (another dial) and steps into it.
Enter submenu	Edits the submenu under the selected direction.
Custom script... / Edit script...	Puts your own MAXScript under the direction, or edits it when the direction already runs a script. It runs with full rights (chapter 8).
Script library...	Puts a ready-made, tested script under the direction (chapter 8).
Value...	Turns the direction into a value slider (chapter 8).
Change label...	Your own caption instead of 3ds Max's name. Shorter is better on the dial.
Settings...	Whether an ordinary release opens this command's caddy: as the global setting, always, or never. Crossing the rim still reverses it. For a script: its second version <i>with parameters</i> (the command with its caddy, say) - one runs on an ordinary release, the other past the rim.
Colour...	This one item's caption and tile colour.
Empty the direction	Removes the item; the direction cancels.
Usage conclusions	Hints drawn from the usage counters, such as a direction you aim at often that is empty. Quick gestures and held dials are both counted. The editor never rearranges anything by itself.
A tap runs	What a press-and-release without movement runs: nothing, or one of the directions.
List under the dial	Up to twelve rows. The first row of buttons fills the list: Add to the list (the command selected in the catalogue), Add a script... , Script library... , Remove row , ▲ ▼ to reorder; commands can also be dragged in. The second row edits the selected row like a direction: Change label... , Edit script... , Settings... , Colour... (the caption only - a row has no tile).

Right - the catalogue

Control	What it does
Search field	Filters 3ds Max's 4000+ actions and macroscripts by name, category or table. The same name can exist in several tables - the category column tells them apart.
Assign to selected direction	Puts the chosen command under the selected direction.

Hover a command	Shows its table and id - useful when writing menus.json by hand.
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Bottom row

Button	What it does
Assign to selected	Same as the catalogue button.
Experimental features...	The toggles of chapter 11, each with its reason.
Colours...	The colour editor (chapter 10).
Clear the counters	Resets the usage statistics. Menus are not changed.
Cheat sheet...	Writes your SAVED dials - with their current keys and mouse bindings, list rows and submenu paths - into one HTML and PDF document to print, sorted by shortcut, and lists what needs checking (dials that would open empty, clashing mouse bindings). Unsaved edits are not included; a warning says so.
Load default dials...	Brings in the built-in 24 dials. By default only slots without their own menu are filled; tick the option to replace the others too. Shows the list before changing.
Save preset...	Saves your dial layout to a file you can share - without usage data or paths.
Load preset...	Loads a layout, replacing the slots it contains. Script and value items are removed by default wherever they sit - directions, list rows, the tap item, submenus and variants - because a preset is somebody else's code; the editor shows the contents first and reports how many dials it loaded.
Reload from file	Discards unsaved changes and reads menus.json again.
Save	Writes menus.json; the plugin reloads it by itself. If menus.json changed on disk after the editor loaded it (another editor, a tick in the Experimental features menu), Save refuses and offers Save your edits as a separate copy ; nothing is overwritten. A file that could not be read must be repaired and reloaded before Save works.
Closing the window	With unsaved changes, Esc or the X asks: Save, Discard or Cancel. Nothing is lost silently.

MarkingForge - dial cheat sheet

2026-10-01 01:46 · 24 dials · sorted by shortcut, because that is what the hand presses

A1t+1 Modelling

slot 1

1. when window under the cursor matches slate

↖ NH Pick from selected	↑ N New material	↗ NE Edit selected's material
← W Get from object	Materials (Slate)	→ E Assign to selection
↙ SW Material Explorer	↓ S Material/Map Browser	↘ SE Show in viewport

2. when window under the cursor matches trackView

↖ NH Curve mode	↑ N Tangents: auto	↗ NE Tangents: smooth
← W Isolate curve	Track View	→ E Tangents: linear
↙ SW Frame all	↓ S Flatten	↘ SE Tangents: stepped

3. when window under the cursor matches particleView

↖ NH Rename	↑ N Move up	↗ NE Copy
← W Move left	Particle View	→ E Move right
↙ SW Cut	↓ S Move down	↘ SE Paste

4. when window under the cursor matches sceneExplorer

↖ NH Zoom to selected	↑ N Select all	↗ NE Select children
← W Unhide all	Scene Explorer	→ E Hide
↙ SW Unfreeze all	↓ S Isolate	↘ SE Freeze

List below the dial

1. Expand all
2. Collapse all
3. Find selected
4. Rename

5. when window under the cursor matches commandPanel

Modifier stack. Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.

6. when modifier = Unwrap_UVW

↖ NH Stitch	↑ N Quick relax	↗ NE Pack
← W Weld selected	UV (Unwrap)	→ E Edge loop
↙ SW Break	↓ S Straighten	↘ SE Edge ring

List below the dial

1. Open UV Editor

7. when modifier = Edit_Poly and subobject: 1 - usually vertex

↖ NH Object level	↑ N Chamfer	↗ NE Weld
← W Target Weld	Edit Poly: vertex	→ E Connect
↙ SW Remove	↓ S Extrude	↘ SE Break

List below the dial

1. Grow
2. Shrink
3. Remove isolated

8. when modifier = Edit_Poly and subobject: 2 - usually edge

↖ NH Object level	↑ N Connect	↗ NE Chamfer
← W Remove	Edit Poly: edge	→ E Loop
↙ SW Weld	↓ S Extrude	↘ SE Ring

List below the dial

1. Split
2. Turn
3. Grow
4. Shrink

9. when modifier = Edit_Poly and subobject: 3 - usually border / spline

↖ NH Object level	↑ N Cap	↗ NE Chamfer
← W Grow	Edit Poly: border	→ E Extrude
↙ SW Shrink	↓ S Create shape	↘ SE Connect

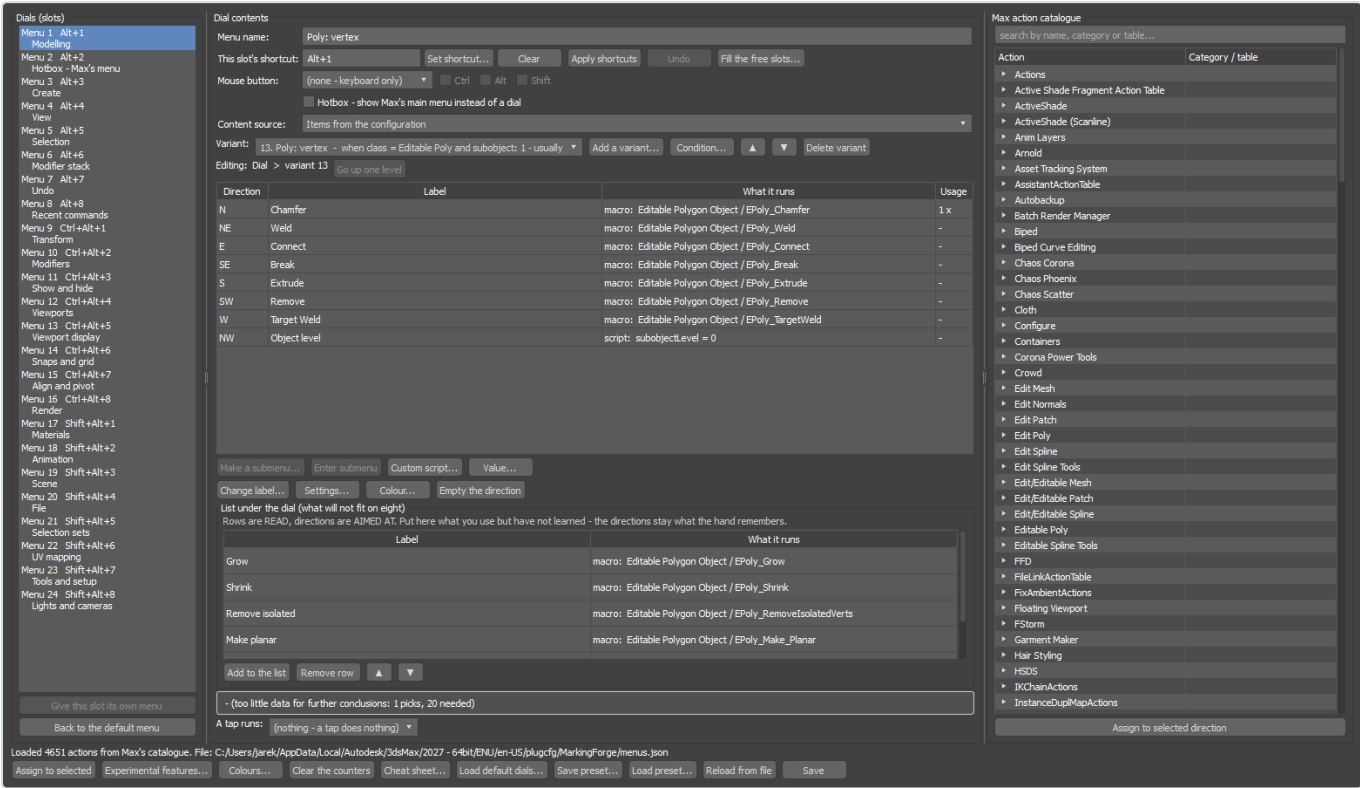
List below the dial

1. Bridge

The first page of the cheat sheet (Cheat sheet...): every dial as a 3x3 square, sorted by shortcut - print it and keep it beside the keyboard for the first week.

6. Contextual variants (conditions)

A variant appears when **all** filled fields match; an empty field is not tested. Variants are tried from the top and the first match wins, so put the specific ones first - for example *Edit Poly at Vertex level* above *Edit Poly*.



The editor with the variant *Poly: vertex* of the Modelling dial selected - what Alt+1 shows on an Editable Poly at vertex level.

The variant appears when ALL the filled-in fields match at once.
A field left empty means 'any' - it is not tested.

Object class:

Editable Poly

Take from selection

Object kind:

(any kind)

Modifier in the panel:

e.g. Bend (empty = do not test)

Has modifier:

e.g. TurboSmooth (empty = do not test)

Sub-object level:

1 - usually vertex

Cursor over the window:

(do not test the window)

What is open?

Selection:

any number (do not test)

0

Condition: when class = Editable Poly and subobject: 1 - usually vertex

With the current selection: does not match (selected: 1 | class: 'Cylinder' | superclass: 'geometry' | subobject: 0 | modifier: '-')

OK

Cancel

Condition... - the fields of a variant's condition. **Take from selection** fills them from the object selected in 3ds Max.

Field	Matches when	Tips
-------	--------------	------

Object class	The selected object's base object is of this class, for example <code>Editable Poly</code> .	Take from selection fills it in the exact spelling. Class looks UNDER the modifiers: an Editable Poly with a TurboSmooth on top is still an Editable Poly. Both spellings (<code>Editable Poly</code> / <code>Editable_Poly</code>) work.
Object kind	The object is of this kind: geometry, shape, light, camera, helper.	Wider than a class - one variant for every light.
Modifier in the panel	The Modify panel is currently editing this modifier, for example <code>Edit_Poly</code> .	Put these ABOVE class variants: an Edit Poly on an Editable Poly matches both.
Has modifier	The object has this modifier anywhere in its stack.	Answers "does it have it", not "is it open".
Sub-object level	object level (0); 1 - usually vertex; 2 - usually edge; 3 - usually border / spline; 4 - usually polygon; 5 - usually element.	The numbers mean vertex/edge/border/polygon/element on a poly and vertex/segment/spline on a shape.
Cursor over the window	The cursor is over that window (table below).	What is open? lists the windows open right now with the word to type.
Selection	nothing selected; exactly one object; two or more; at least one, or an exact number.	<i>nothing selected</i> gives an empty scene its own menu.
Name	The object's name matches a pattern with <code>*</code> , for example <code>Rig_*</code> .	Handy for rigs and props named by convention (menus.json field <code>name</code>).
Layer	The object is on this layer.	menus.json field <code>layer</code> .

Windows the plugin recognises

Word	Window	Details
(empty)	(do not test the window)	The rule ignores where the cursor is.
<code>viewport</code>	3D view	Decided by Max's SDK, not by a window name - so it works regardless of window layout and of language.
<code>commandPanel</code>	Command panel	The right-hand panel with the Create / Modify / Hierarchy tabs.
<code>slate</code>	Material editor (Slate)	The Slate Material Editor, together with its child windows.
<code>materialBrowser</code>	Material/Map Browser	The material and map list inside Slate. NARROWER than 'slate', which also covers it - use this one when the rule is about the browser itself.
<code>materialParams</code>	Material parameters (Slate)	The parameter panel of the material under the cursor, inside Slate. Also narrower than 'slate'.
<code>trackView</code>	Track View	The curve and key editor. Recognised by object name, because its class is shared by many docked windows. MEASURED: the object is called 'TrackView - Track View - Curve Editor', the class is <code>MaxSDK::QmaxDockWidget</code> .
<code>sceneExplorer</code>	Scene Explorer	The scene object list. The same generic class as Track View, so it is likewise decided by object name: 'Scene Explorer - <name>'.
<code>particleView</code>	Particle View	The ParticleFlow window. The ONLY one on this list that is not a Qt window at all but an old Windows dialog of class 'Particle View Dialog'. On the Qt side it belongs to the main window, so without this word it could not be told apart from it.

ribbon	Ribbon (modelling toolbar)	The ribbon with the modelling tools. MEASURED: object name 'Ribbon', class MaxSDK::QmaxDockWidget - the generic docked class, so the name decides.
timeSlider	Time slider	The animation bar under the views. MEASURED: object name 'TimeSlider', class QmaxTimeSliderDockWidget.
floatingViewport	Floating view	A view torn off into a window of its own. NOT the same as '3D view': that one is decided by the SDK against the main window's views, while a torn-off panel is a separate window.
mainWindow	Max's main window	The main window's OWN surface - toolbars, time slider, background. It does NOT cover what Max hosts inside it: neither the 3D view nor old dialogs such as Particle View. Until 0.39.1 it did, and that was wrong: the 'main window' rule fired over half of Max.

With the experimental *Menu under the cursor* on (default), the object UNDER the cursor decides the variant when there is one; otherwise the selection does. Picking an item then selects that object and runs the command on it in one undo. Hotbox entries and type-to-search results run without changing the selection.

7. Content sources

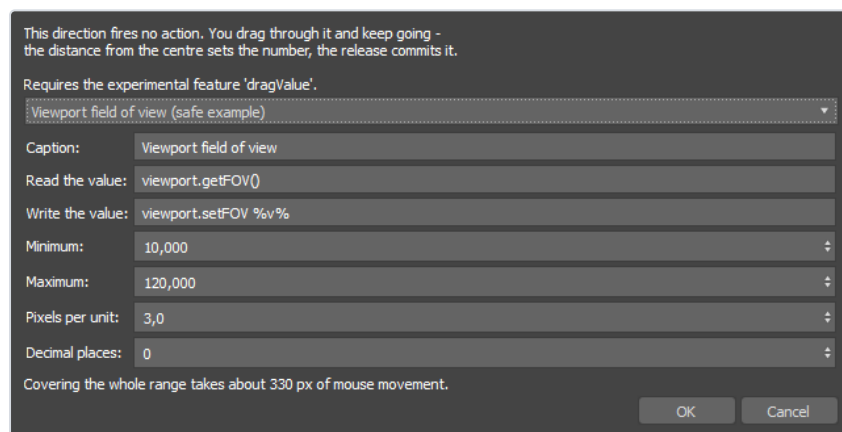
Source	Word in menus.json	What the dial shows
Items from the configuration	<code>config</code>	Eight directions exactly as you set them here. An ordinary dial.
Modifier stack	<code>modifiers</code>	Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.
Undo stack	<code>undo</code>	The last eight operations BY NAME. Direction N undoes one, NE two, and so on. Requires the experimental feature 'undoDial'.
Recent commands	<code>recent</code>	The eight most recently RUN commands, newest first - from any dial, a tap or the search. For REPEATING, as opposed to the undo stack. The history lives in memory and starts afresh when Max starts. Requires the 'recentDial' toggle.
Selection sets	<code>selectionSets</code>	The scene's named selection sets as eight directions. With more than eight it shows the first eight. Requires the toggle 'selectionSetsDial'.

A source can be set on the whole dial or on one variant only - for example the modifier stack only while the cursor is over the command panel (the built-in Modelling dial does exactly this). Items typed into a dial with a source are kept, not lost; they come back when you switch the source to *Items from the configuration*.

8. Item types

Type	Runs	Fields in menus.json
Action	A 3ds Max action, from any action table - built-in or another plugin's.	<code>table</code> (number), <code>id</code> (text), optional <code>label</code> , <code>settings</code>
Macro	A macroscript.	<code>category</code> , <code>name</code> , optional <code>label</code> , <code>settings</code>
Script	Your own MAXScript code, with full rights.	<code>code</code> , optional <code>label</code> , <code>codeSettings</code> (code for the settings variant of the gesture)
Submenu	Another dial, opened by moving through the direction. Up to three levels.	<code>label</code> , <code>items</code> (eight directions), optional <code>list</code>
Value	A slider: the distance from the centre sets a number, the release commits it. Needs <i>dragValue</i> .	<code>get</code> (expression returning the start value), <code>set</code> (code containing <code>%v%</code>), <code>min</code> , <code>max</code> , <code>pxPerUnit</code> , <code>decimals</code> (0-4)

The value slider in the editor



Value... with its first ready-made example (viewport field of view).

Field	Meaning
Caption	The dial caption; the number appears next to it.
Read the value	A MAXScript expression returning the starting number, read when the cursor enters the direction. No number (nothing selected) shows <code>?</code> .
Write the value	Code run on release. Must contain <code>%v%</code> , which becomes the value.
Minimum / Maximum	The range the gesture cannot leave.
Pixels per unit	Mouse movement per one unit. Smaller = more sensitive. For 0-4 use about 40, for 0-500 about 2.
Decimal places	0 = whole numbers.
Example	Ready-made examples; the first (viewport field of view) needs no selection.

```
{ "type": "value", "label": "FOV",  
  "get": "viewport.getFOV()",  
  "set": "viewport.setFOV %v%",  
  "min": 10, "max": 120, "pxPerUnit": 3, "decimals": 0 }
```

Scripts: ready-made or your own

A script direction runs MAXScript with full rights. There are two ways to get one, and both put the same kind of item on the dial:

Button	What you get
Script library...	One of 25 ready-made scripts. Each was run in 3ds Max on a test scene before it went into the list. Pick one - it arrives with its caption.
Custom script...	Your own code - typed, pasted from the Listener or from a script you already use.
Edit script...	Shown instead of <i>Custom script...</i> when the direction already runs a script: opens its code. The list under the dial has the same three buttons.

The script library

Group	Caption	What it does
Objects and pivots	Pivot to bottom	Moves each selected object's pivot to the centre of its bottom face - so it sits on the grid and scales from the floor.
Objects and pivots	Drop to the ground	Moves each selected object straight down (or up) until its lowest point is at Z = 0.
Objects and pivots	Centre pivot	Puts each selected object's pivot in the centre of its bounding box.
Objects and pivots	Reset XForm + collapse	Bakes rotation and scale into the mesh and clears the stack - the usual clean-up before exporting or before modifiers that misbehave on a scaled object.
Objects and pivots	Clone in place	Makes an independent copy of the selection on the same spot and selects the copies - Shift-drag without the drag.
Objects and pivots	Make unique	Turns instances and references among the selection into independent copies.
Topology (Editable Poly)	Select n-gons	Selects every polygon with five or more sides on the selected Editable Poly - the quickest topology check before subdividing.
Topology (Editable Poly)	Select triangles	Selects every three-sided polygon on the selected Editable Poly.
Topology (Editable Poly)	Select holes	Selects the open edges (borders) of the selected Editable Poly - every hole in the mesh.
Topology (Editable Poly)	Weld close vertices	Welds vertices closer than 0.01 units on every selected Editable Poly - cleans up doubled vertices after an import or a mirror.
Topology (Editable Poly)	Auto smooth 30	Sets smoothing groups by a 30-degree angle on every selected Editable Poly.
Modifiers	Smoothing on/off	Switches every TurboSmooth and MeshSmooth on the selected objects on or off - a fast low-poly / high-poly check.
Modifiers	+ UVW Map (box)	Adds a box-projection UVW Map to every selected object.
Modifiers	Remove top modifier	Deletes the topmost modifier of every selected object.
Modifiers	Extrude (Edit Poly)	For an Edit Poly modifier at polygon level: an ordinary release starts an interactive Extrude, past the rim opens the Extrude caddy instead. (<i>two versions - see below</i>)

Materials and display	Grey clay material	Gives the selection one neutral grey material - for checking forms and lighting without textures.
Materials and display	Remove material	Clears the material from every selected object.
Materials and display	Random wire colour	Gives every selected object its own random wire colour - tells parts apart in a crowded scene.
Materials and display	See-through on/off	Switches See-Through display (Alt+X) for each selected object.
Materials and display	Display as box on/off	Shows each selected object as its bounding box, or back - lightens heavy objects in the viewport.
Selection and transforms	Select same material	Selects every object that has the same material as the selected one.
Selection and transforms	Select instances	Adds every instance of the selected objects to the selection.
Selection and transforms	Copy transform	Remembers the selected object's position, rotation and scale (pair it with Paste transform).
Selection and transforms	Paste transform	Gives every selected object the transform remembered by Copy transform.
Selection and transforms	Reset rotation	Removes the rotation of each selected object and keeps it where it is.

Four scripts worth having on a dial

The same four are in the library; the code is printed here so you can see what runs and adapt it.

1. Pivot to bottom

Moves each selected object's pivot to the centre of its bottom face - so it sits on the grid and scales from the floor. Objects sit on the grid and scale from the floor - furniture, props, anything placed on a surface.

```
(
  local picked = selection as array
  if picked.count == 0 then
    format "MarkingForge: nothing selected\n"
  else
    for o in picked where (isValidNode o) do o.pivot = [o.center.x, o.center.y, o.min.z]
)
```

2. Select n-gons

Selects every polygon with five or more sides on the selected Editable Poly - the quickest topology check before subdividing. Before TurboSmooth or export: every polygon with five or more sides lights up, and the Listener says how many.

```
(
  local o = selection[1]
  if selection.count != 1 or (classOf o.baseObject) != Editable_Poly then
    format "MarkingForge: select ONE Editable Poly object\n"
  else
    (
      local found = for f = 1 to (polyop.getNumFaces o) where ((polyop.getFaceDeg o f) > 4) collect
      f
    )
  )
```

```

max modify mode
modPanel.setCurrentObject o.baseObject
subobjectLevel = 4
polyop.setFaceSelection o (found as bitArray)
redrawViews()
format "MarkingForge: % polygon(s) with 5+ sides\n" found.count
)
)

```

3. Reset XForm + collapse

Bakes rotation and scale into the mesh and clears the stack - the usual clean-up before exporting or before modifiers that misbehave on a scaled object. A scaled or rotated object misbehaves under Shell, Symmetry or Bevel, and exports with a transform the next program reads differently. This bakes it into the mesh.

```

(
  local picked = selection as array
  if picked.count == 0 then
    format "MarkingForge: nothing selected\n"
  else
    for o in picked where (isValidNode o) and (superClassOf o == GeometryClass) do (ResetXForm o;
maxOps.CollapseNode o off)
)

```

4. Smoothing on/off

Switches every TurboSmooth and MeshSmooth on the selected objects on or off - a fast low-poly / high-poly check. Flip between the cage and the smoothed result on every selected object at once - the low-poly / high-poly check you do a hundred times a day.

```

(
  local picked = selection as array
  if picked.count == 0 then
    format "MarkingForge: nothing selected\n"
  else
    for o in picked where (isValidNode o) do (for m in o.modifiers where (classOf m ==
TurboSmooth or classOf m == MeshSmooth) do m.enabled = not m.enabled)
)

```

From the library: pick the dial, select a direction, **Script library...**, choose the script, **Put on direction**, **Save**. **Your own code:** select a direction, **Custom script...**, paste, OK, **Change label...** for a caption, **Save**.

A script with two versions (the rim)

A command with a caddy opens it on an ordinary release and skips it past the rim (or the other way round, chapter 2). A script can do the same: **Settings...** on a script item takes a second version *with parameters* - here the Extrude caddy of an Edit Poly modifier. Which version an ordinary release runs is chosen in the same window; releasing past the rim runs the other one.

Settings choice	Ordinary release	Past the rim
As the global setting (default: with parameters)	the version with parameters	the plain code
Always with parameters	the version with parameters	the plain code
Always without parameters	the plain code	the version with parameters

The library's **Extrude (Edit Poly)** is set up as *Always without parameters*: plain Extrude inside, the caddy past the rim.

```
// the plain code
(
  local m = modPanel.getCurrentObject()
  if classOf m != Edit_Poly then
    format "MarkingForge: no Edit Poly in the Modify panel\n"
  else
    m.EnterCommandMode #ExtrudeFace
)

// the version with parameters
(
  local m = modPanel.getCurrentObject()
  if classOf m != Edit_Poly then
    format "MarkingForge: no Edit Poly in the Modify panel\n"
  else
    m.PopupDialog #ExtrudeFace
)
```

Writing your own - the shape that works

Every library script has the same three habits; copy them and a script behaves like a built-in item:

- **Snapshot the selection first** - `selection as array`. A command that creates or deletes objects would otherwise change the very set being walked.
- **Say why nothing happened.** With nothing selected, print a line to the MAXScript Listener instead of failing silently.
- **Skip what cannot take the operation** - `where (isValidNode o)`, plus a class test where it matters - instead of stopping on the first light or camera.

```
(
  local picked = selection as array
  if picked.count == 0 then
    format "MarkingForge: nothing selected\n"
  else
    for o in picked where (isValidNode o) do ( /* your code - o is each selected object */ )
)
```

Scripts run with full rights, like an .mcr file. Paste only code you understand, and never code from a scene or a preset you have not read (chapter 13).

9. Shortcuts and mouse buttons

Keyboard

The standard scheme is Alt+1..8 (dials 1-8), Ctrl+Alt+1..8 (9-16), Shift+Alt+1..8 (17-24). Three ways to assign keys:

- **MarkingForge > Set up standard shortcuts...** - free dials only, skips keys already in use.
- The editor: *This slot's shortcut* + **Apply shortcuts**, or **Fill the free slots...**
- 3ds Max's **Customize > Hotkey Editor**: the actions *MarkingForge: Menu 1..24*.

MarkingForge keeps its shortcuts in its own hotkey set, `MarkingForge_Hotkeys`, which contains all of your other shortcuts too. Every change is preceded by a backup in `plugcfg\MarkingForge\hotkeys\backup_*.hsx`.

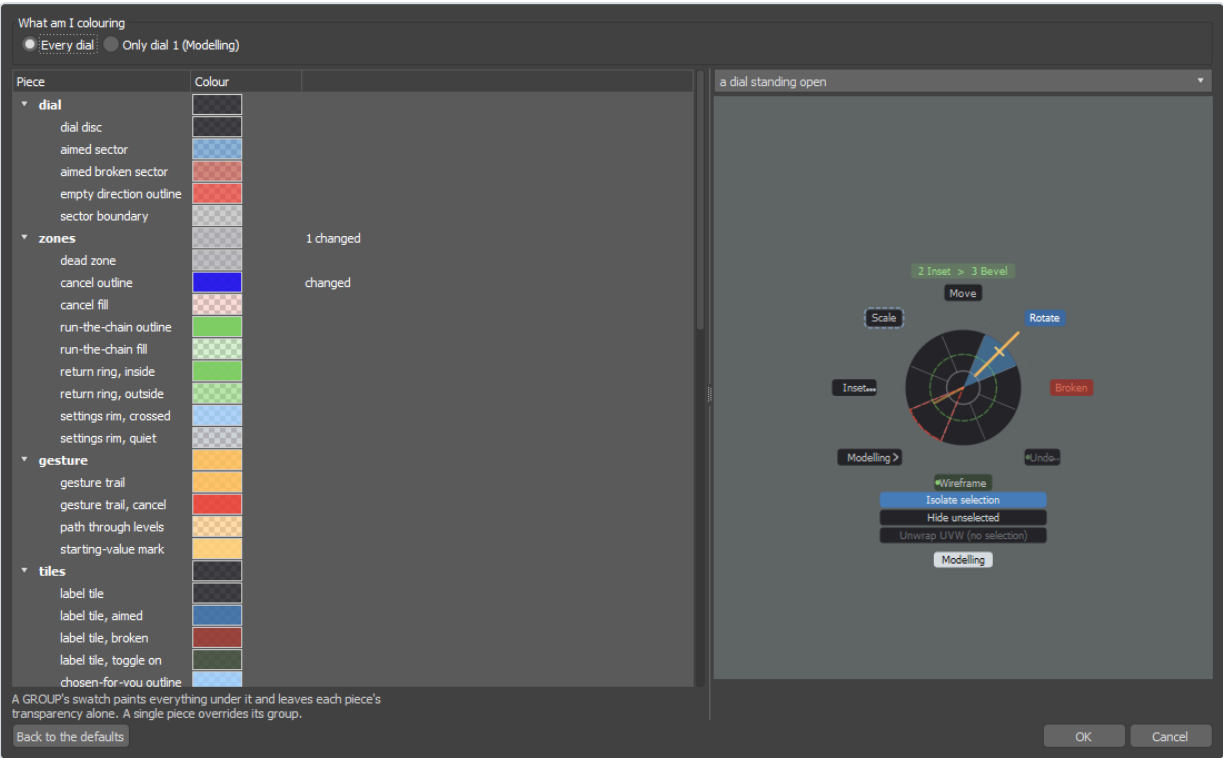
Mouse buttons

Button	Name in menus.json
(none - keyboard only)	
left button	LMB
middle button (wheel)	MMB
right button	RMB
thumb button 1 (back)	XB1
thumb button 2 (forward)	XB2

Modifiers: Ctrl, Alt, Shift - at least one is required, for example `"mouse": "Ctrl+Shift+MMB"`. A bare button would be taken from 3ds Max (left selects, middle pans, right opens the quad menu), so both the editor and the plugin refuse it. 3ds Max also uses some modifier+button combinations for navigation - if one collides with a habit of yours, pick another. With no dial bound to a button the plugin installs no mouse hook at all.

10. Colours

Colours... in the editor. Two levels: a **group** colours everything in it (keeping each piece's transparency), a **single piece** overrides its group. **Scope** chooses all dials or this dial only - a different colour per dial is the quickest way to know which one opened. **Colour...** under the directions colours one item's caption and tile. Only deviations from the defaults are saved.



The colour editor with its live preview.

```
"theme": { "groups": { "dial": "#101820", "tiles": "#1c3a20" },
  "colors": { "tileFillSelected": "#3f8a4aff" } }
```

Colours are `#RRGGBB` or `#RRGGBBAA` (AA = opacity). The same `theme` can sit at the top of `menus.json` (all dials) or inside one menu.

Group	Name	Piece	Default
dial	dialFill	dial disc	#18181acd
dial	sectorFill	aimed sector	#508cc8a5
dial	sectorBroken	aimed broken sector	#be463ca5
dial	sectorEmpty	empty direction outline	#eb463cc8
dial	spoke	sector boundary	#96969b78
zones	deadZone	dead zone	#96969b96
zones	cancelRing	cancel outline	#eb6e5aeb
zones	cancelFill	cancel fill	#eb6e5a37
zones	queueRing	run-the-chain outline	#78cd5ff0
zones	queueFill	run-the-chain fill	#78cd5f46
zones	returnRingNear	return ring, inside	#78cd5feb

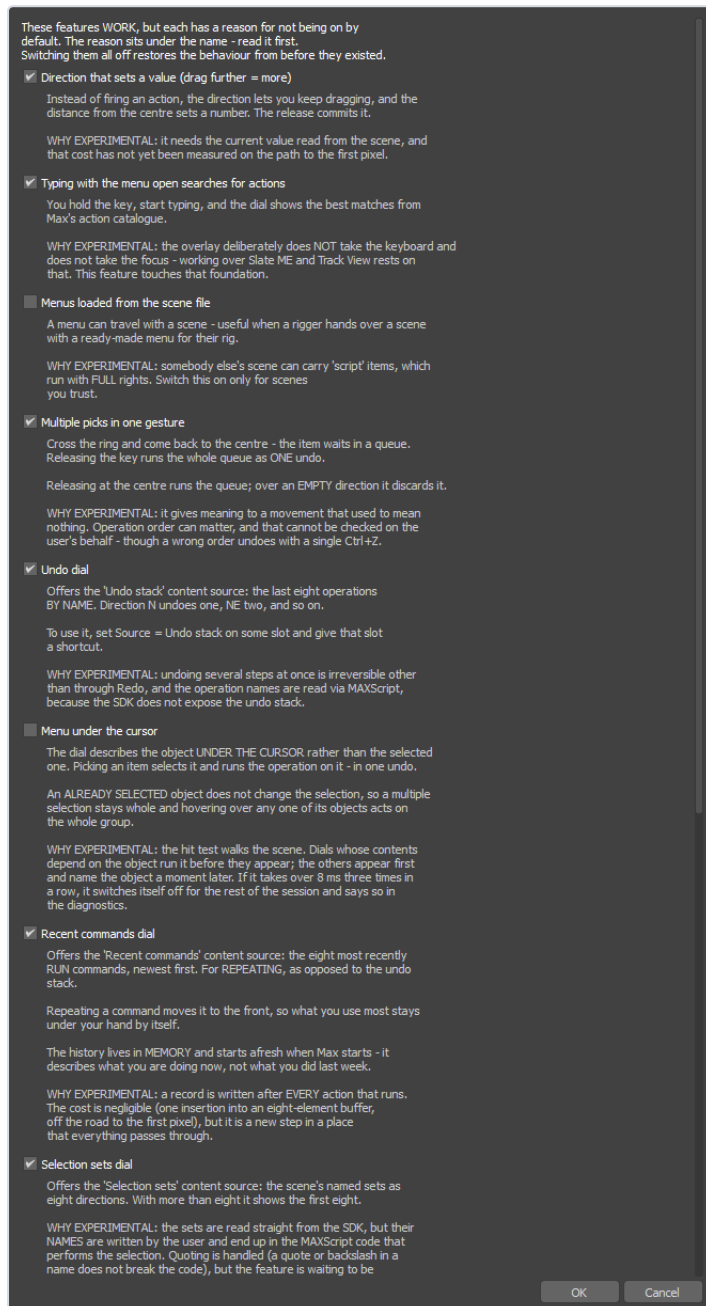
zones	<code>returnRingFar</code>	return ring, outside	 #78cd5f82
zones	<code>settingsRingActive</code>	settings rim, crossed	 #96c8fabe
zones	<code>settingsRingIdle</code>	settings rim, quiet	 #7882915a
gesture	<code>trail</code>	gesture trail	 #ffbe5ae1
gesture	<code>trailCancel</code>	gesture trail, cancel	 #eb463cf0
gesture	<code>trailBack</code>	path through levels	 #ffbe5a82
gesture	<code>valueMark</code>	starting-value mark	 #ffd278e6
tiles	<code>tileFill</code>	label tile	 #18181acd
tiles	<code>tileFillSelected</code>	label tile, aimed	 #3c6ea5eb
tiles	<code>tileFillBroken</code>	label tile, broken	 #963730eb
tiles	<code>tileFillChecked</code>	label tile, toggle on	 #34402ed7
tiles	<code>tileDefaultOutline</code>	chosen-for-you outline	 #96c8fad2
tiles	<code>tileQueuedDot</code>	already-queued dot	 #78cd5fe6
tiles	<code>tileCheckDot</code>	toggle dot	 #96d778f0
tiles	<code>tileCheckDotDim</code>	toggle dot, unavailable	 #6e8c64c8
tiles	<code>tileArrow</code>	submenu arrow	 #c8c8d2be
tiles	<code>tileArrowSelected</code>	submenu arrow, aimed	 #ffffffeb
tiles	<code>tileEllipsis</code>	settings dots	 #d7e1f0d7
tiles	<code>tileEllipsisSelected</code>	settings dots, aimed	 #fffffff5
tiles	<code>tileEllipsisIdle</code>	settings dots, dimmed	 #969eac6e
tiles	<code>tileEllipsisIdleSel</code>	dimmed dots, aimed	 #ffffff69
text	<code>textLabel</code>	label	 #d7d7ded7
text	<code>textLabelSelected</code>	label, aimed	 #ffffffff
text	<code>textBroken</code>	label, broken	 #eb7864cd
text	<code>textBrokenSelected</code>	label, broken and aimed	 #ffebe6ff
text	<code>textDim</code>	label, unavailable	 #828288be
text	<code>textDimSelected</code>	unavailable and aimed	 #aaaaafeb
list	<code>listFill</code>	list row	 #18181acd
list	<code>listFillPointed</code>	list row, pointed at	 #4682c8e1
list	<code>listText</code>	list row text	 #e8ecf2d7
list	<code>listTextPointed</code>	list text, pointed at	 #e8ecf2ff
list	<code>listTextDim</code>	list text, unavailable	 #96969ba0
footer	<code>footerTile</code>	footer tile	 #e8ecf2e1
footer	<code>footerText</code>	footer text	 #000000eb
chain	<code>chainTile</code>	chain tile	 #78cd5f37
chain	<code>chainText</code>	chain text	 #a0e68cf0

hotbox	hotboxCentre	hotbox centre dot	 #c8c8d278
hotbox	hotboxTitleFill	menu title tile	 #18181ad2
hotbox	hotboxTitleFillOpen	menu title, open	 #3c6ea5eb
hotbox	hotboxTitleText	menu title text	 #d7d7ded7
hotbox	hotboxTitleTextOpen	menu title text, open	 #fffffffa
hotbox	hotboxPanel	expanded list panel	 #121214e6
hotbox	hotboxRowFillHot	hotbox row, pointed at	 #3c6ea5eb
hotbox	hotboxRowText	hotbox row text	 #d2d2dad2
hotbox	hotboxRowTextHot	hotbox row text, hot	 #fffffffa

11. Experimental features

Features that work but each have a reason not to be forced on everyone. Switch them in the editor (**Experimental features...**) or in **MarkingForge > Experimental features**; the tick in the menu shows the state. Switching all of them off restores the behaviour from before they existed.

A tick in the MarkingForge menu saves `menus.json` at once. An editor that is open at that moment then refuses to Save until you reload (see Save in chapter 7).



The experimental features dialog.

Direction that sets a value (drag further = more) `dragValue` OFF BY DEFAULT

Instead of firing an action, the direction lets you keep dragging, and the distance from the centre sets a number. The release commits it.

WHY EXPERIMENTAL: it needs the current value read from the scene, and that cost has not yet been measured on the path to the first pixel.

Typing with the menu open searches for actions `gestureSearch` OFF BY DEFAULT

You hold the key, start typing, and the dial shows the best matches from Max's action catalogue.

WHY EXPERIMENTAL: the overlay deliberately does NOT take the keyboard and does not take the focus - working over Slate ME and Track View rests on that. This feature touches that foundation.

Menus loaded from the scene file `sceneMenus` OFF BY DEFAULT

A menu can travel with a scene - useful when a rigger hands over a scene with a ready-made menu for their rig.

WHY EXPERIMENTAL: somebody else's scene can carry 'script' items, which run with FULL rights. Switch this on only for scenes you trust.

Multiple picks in one gesture `multiPick` OFF BY DEFAULT

Cross the ring and come back to the dashed circle round the centre - the item waits in a queue. Releasing in that circle (it is green then) runs the whole queue as ONE undo.

To give up: release at the very CENTRE (the small dot) or over an EMPTY direction - the queue is discarded.

WHY EXPERIMENTAL AND OFF: it gives meaning to a movement that used to mean nothing - going a little past the ring and coming back is also how a hand gives up on a direction. Operation order can matter too - though a wrong order undoes with a single Ctrl+Z.

Undo dial `undoDial` ON BY DEFAULT

Offers the 'Undo stack' content source: the last eight operations BY NAME. Direction N undoes one, NE two, and so on.

To use it, set Source = Undo stack on some slot and give that slot a shortcut.

WHY EXPERIMENTAL: undoing several steps at once is irreversible other than through Redo, and the operation names are read via MAXScript, because the SDK does not expose the undo stack.

Menu under the cursor `hoverTarget` ON BY DEFAULT

The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects it and runs the operation on it - in one undo.

An ALREADY SELECTED object does not change the selection, so a multiple selection stays whole and hovering over any one of its objects acts on the whole group.

WHY EXPERIMENTAL: the hit test walks the scene. Dials whose contents depend on the object run it before they appear; the others appear first and name the object a moment later. If it takes over 8 ms three times in a row, it switches itself off for the rest of the session and says so in the diagnostics.

Recent commands dial recentDial **ON BY DEFAULT**

Offers the 'Recent commands' content source: the eight most recently RUN commands, newest first. For REPEATING, as opposed to the undo stack.

Repeating a command moves it to the front, so what you use most stays under your hand by itself.

The history lives in MEMORY and starts afresh when Max starts - it describes what you are doing now, not what you did last week.

WHY EXPERIMENTAL: a record is written after EVERY action that runs. The cost is negligible (one insertion into an eight-element buffer, off the road to the first pixel), but it is a new step in a place that everything passes through.

Selection sets dial selectionSetsDial **ON BY DEFAULT**

Offers the 'Selection sets' content source: the scene's named sets as eight directions. With more than eight it shows the first eight.

WHY EXPERIMENTAL: the sets are read straight from the SDK, but their NAMES are written by the user and end up in the MAXScript code that performs the selection. Quoting is handled (a quote or backslash in a name does not break the code), but the feature is waiting to be confirmed in real work.

Modifier under the cursor modifierUnderCursor **ON BY DEFAULT**

A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from.

WHY EXPERIMENTAL: the same risk as the menu under the cursor - the dial talks about an object the user did not select, so that must be visible. The object's name is in the dial header, marked [under cursor].

Operation-pair learning pairLearning **ON BY DEFAULT**

The plugin remembers what follows what and sets the dial's DEFAULT direction to the most frequent successor of the last operation - the one a TAP of the key will run (and Enter, when the dial is pinned). A pair counts from its second repetition; a 'tap' item of the dial always wins over the suggestion.

WHAT IT DOES NOT DO: it does not move items. A menu that reorders itself turns a remembered gesture into reading, and muscle memory is the whole point of a marking menu.

WHY EXPERIMENTAL: the memory is SESSION-ONLY and nobody has yet measured whether the hints land. The hit counter is in 'Configuration status'.

Chain as an item chainAsItem **ON BY DEFAULT**

After you run a chain (multiple picks) the plugin REMEMBERS it. You then store it with 'Save last chain...' in the MarkingForge menu.

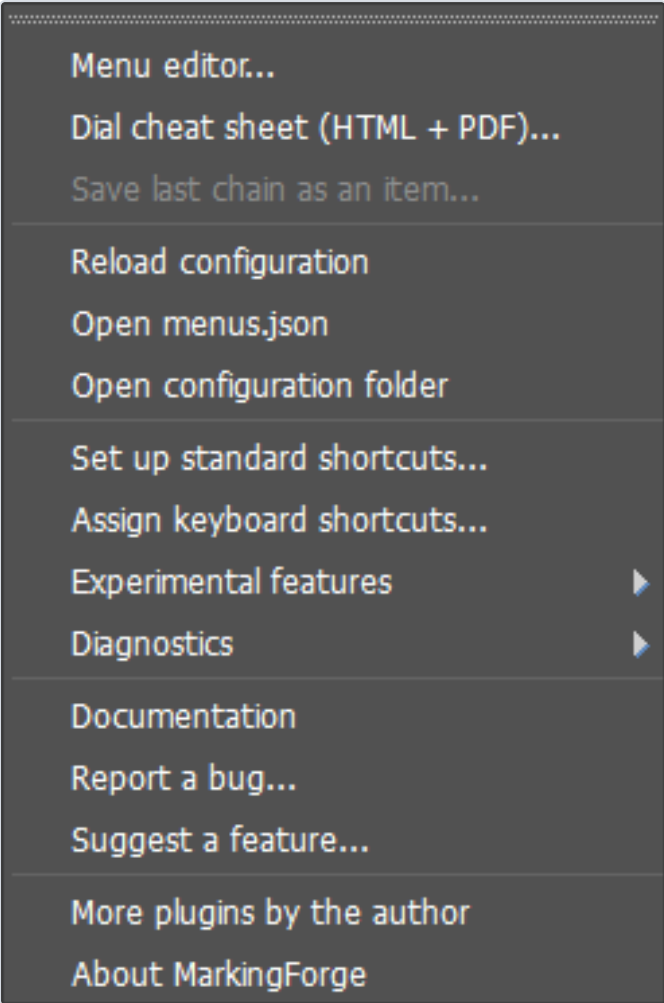
DELIBERATELY NO PROMPT AFTER EVERY GESTURE: a dialog popping up after a chain would turn a feature that speeds work up into one that interrupts it.

WHY EXPERIMENTAL: the result is stored as a 'script' item, and replay works only for actions and macros - a chain containing a 'value' item or a submenu will not save at all, and says so.

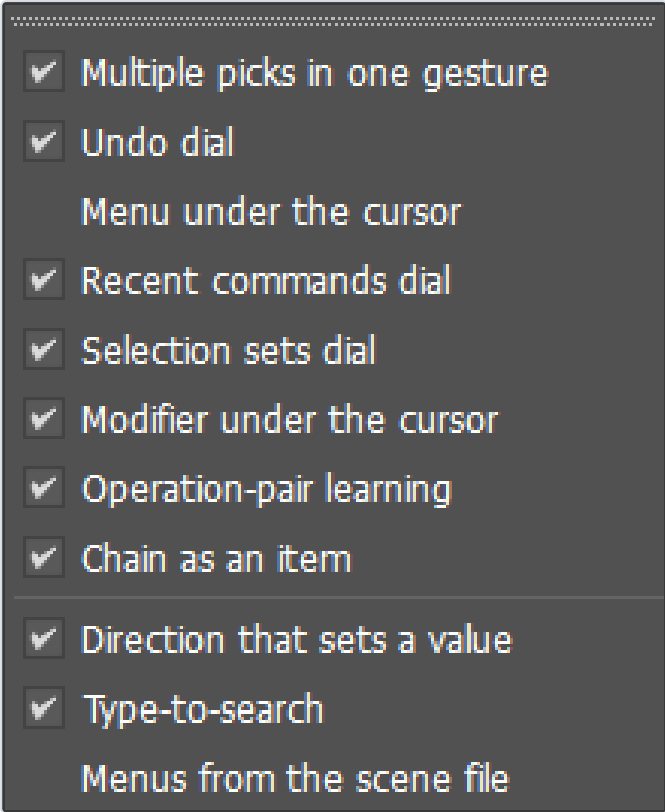
An ordinary release shows the action's settings showItemSettings **ON BY DEFAULT**

Picking Inset, Extrude, Bevel... runs the command and opens its caddy, as the quad menu's settings icon does. This is only the default: crossing the rim reverses it, and **Settings...** can change it per item.

12. The MarkingForge menu



The menu.



Experimental features.

Every command is also an action in the category **MarkingForge**, so it can go on a toolbar, into a menu or quad menu (**Customize > Menu Editor**), or get a hotkey.

Command	What it does
Menu editor...	Opens a window where you assign actions to dial directions and set keyboard shortcuts.
Dial cheat sheet (HTML + PDF)...	Collects every dial into one document to read and to print. Along the way it checks which dial would open empty.
Save last chain as an item...	Takes the last chain you ran (multiple picks) and stores it as ONE direction in a dial of your choice.
Reload configuration	Reads menus.json again. Useful when the file was edited outside Max.
Open menus.json	Opens the configuration file in your default text editor. The plugin picks up changes by itself.
Open configuration folder	Opens the folder holding menus.json, the shortcut backups and the settings in Explorer.
Assign keyboard shortcuts...	Opens 3ds Max's Hotkey Editor. The dials are the actions 'MarkingForge: Menu 1' ... 'Menu 24' - type MarkingForge into its search field.
Set up standard shortcuts...	Gives every dial without a shortcut its standard one: Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8. Keys already in use are skipped; your own shortcuts are not touched.

Configuration status	Shows which file is loaded, how many menus it holds, whether it has errors, and the state of the experimental features (including how often pair-learning suggestions were taken).
About MarkingForge	Shows the plugin's version, its configuration file, and links to support and to more plugins by the author.
Documentation	Opens the folder with the MarkingForge guides (QuickStart, Reference, Installation and more).
Report a bug...	Copies your 3ds Max and MarkingForge details to the clipboard and opens an e-mail to forgeplugins@gmail.com .
Suggest a feature...	Send an idea or a feature request to forgeplugins@gmail.com .
More plugins by the author	Opens https://forgeplugins.tech in your web browser.
Multiple picks in one gesture	Cross the ring and come back to the centre - the item waits in a queue. The release runs the whole queue as ONE undo.
Undo dial	Offers the "undo" menu source: the last eight operations by name. Assign it to a slot in the menu editor.
Menu under the cursor	The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects that object and runs the operation on it - in a single undo.
Direction that sets a value	A "value" direction turns into a slider - the gesture radius sets the number.
Type-to-search	Typing while the dial is open turns it into a search over Max's actions.
Menus from the scene file	CAUTION: menus carried in by somebody else's scene. Script items are rejected, but switch this on deliberately all the same.
Modifier under the cursor	A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from.
Operation-pair learning	The plugin remembers what follows what and sets the dial's DEFAULT direction to the most frequent successor. It does NOT move items around - muscle memory rests on that. Session memory only.
Chain as an item	After you run a chain (multiple picks) the plugin remembers it. You then store it with 'Save last chain...' - with no dialog popping up after every gesture.
Recent commands dial	Offers the 'Recent commands' content source: the last eight you ran, newest first. For REPEATING, as opposed to the undo stack.
Selection sets dial	Offers the 'Selection sets' content source: the scene's named sets as eight directions.

Diagnostics (event log, configuration status) are in **MarkingForge > Diagnostics**; they print to the MAXScript Listener and change nothing.

13. Menus carried by scenes

A scene file can carry its own dials - for example a rigger handing over a character with a ready menu for its controls. This needs the experimental **sceneMenus** (off by default).

Mode (sceneMenuMode)	Behaviour
Ask - load them, but do NOT use them without my consent (recommended)	The scene's menus are loaded and counted, but the dials STILL show your configuration. The Listener reports that the scene carries something, and you switch with the button below or the sceneMenuUse command.
Always use the scene's menus when it carries any	For a studio where project scenes ARE meant to bring their own menus. CAUTION: this means opening somebody else's file REPLACES your dials without asking. Script items are rejected regardless.
Never - do not even look inside scenes	The plugin does not read menu data from scene files at all. The most cautious setting.

Command	What it does
MarkingForge.sceneMenuSave <slot>	Stores a copy of that slot's menu in the open scene. Save the scene to keep it.
MarkingForge.sceneMenuInfo()	Says whether the scene carries menus and whether they are in use.
MarkingForge.sceneMenuUse true false	Switches between the scene's menus and yours.
MarkingForge.sceneMenuClear()	Removes the menus from the scene.

Script items in a scene's menus are always rejected - somebody else's scene must not run code with your rights.

14. The configuration file (menus.json)

MarkingForge > Open menus.json. Everything the editor does is stored here, and the file can be edited by hand: the plugin reloads it when it is saved. A mistake disables only the item it is in and is reported by **Diagnostics > Configuration status** - the rest keeps working.

Top level

Key	Meaning
<code>schema</code>	File format version. Leave as written.
<code>menus</code>	An object with keys <code>"1"</code> ... <code>"24"</code> - each a menu.
<code>defaultMenu</code>	The menu shown by slots without their own.
<code>experimental</code>	The experimental switches that differ from the defaults, e.g. <code>{ "gestureSearch": true }</code> .
<code>theme</code>	Colours for all dials (chapter 10).
<code>showItemSettings</code>	<code>false</code> = an ordinary release skips the caddy.
<code>sceneMenuMode</code>	<code>ask</code> (default), <code>always</code> or <code>never</code> .

A menu

Key	Meaning
<code>name</code>	The dial's name.
<code>items</code>	Directions <code>N NE E SE S SW W NW</code> , each an item (chapter 8). A missing direction is empty.
<code>list</code>	An array of up to twelve items drawn under the dial.
<code>rules</code>	An array of variants: <code>{ "when": { ... }, "menu": { ... } }</code> . First match wins.
<code>source</code>	<code>modifiers</code> , <code>undo</code> , <code>recent</code> or <code>selectionSets</code> .
<code>layout</code>	<code>"hotbox"</code> for a hotbox.
<code>tap</code>	A direction name (<code>"N"</code>) or a whole item run on a tap.
<code>mouse</code>	A mouse binding, e.g. <code>"Alt+MMB"</code> .
<code>theme</code>	Colours for this dial only.

Conditions (when)

Key	Example
<code>class</code>	<code>"Editable Poly"</code>
<code>superClass</code>	<code>"shape"</code> , <code>"light"</code> , <code>"camera"</code> , <code>"geometry"</code> , <code>"helper"</code>
<code>modifier</code>	<code>"Edit_Poly"</code> (open in the Modify panel)
<code>hasModifier</code>	<code>"TurboSmooth"</code> (anywhere in the stack)
<code>subobject</code>	<code>1</code> ... <code>5</code> , <code>0</code> = object level
<code>window</code>	<code>"slate"</code> , <code>"trackView"</code> ... (chapter 6)
<code>selection</code>	<code>"none"</code> , <code>"one"</code> , <code>"many"</code> , <code>"any"</code> or a number
<code>name</code>	<code>"Rig_*</code>

layer	"Characters"
-------	--------------

Item fields common to all types

Key	Meaning
type	action, macro, script, submenu, value
label	Caption on the dial (optional for action and macro).
settings	true / false - open the caddy on an ordinary release.
codeSettings	Script only: the version <i>with parameters</i> , run instead of <code>code</code> whenever the item wants its parameters (see <code>settings</code> ; reversed past the rim).
color, tile	This item's caption and tile colour.

A complete small example

```
{
  "menus": {
    "3": {
      "name": "My tools",
      "items": {
        "N": { "type": "action", "table": 0, "id": "40021", "label": "Select all" },
        "S": { "type": "action", "table": 0, "id": "40043", "label": "Select none" },
        "E": { "type": "macro", "category": "Modifiers", "name": "TurboSmoothMod" },
        "W": { "type": "script", "label": "Box", "code": "startObjectCreation Box" }
      },
      "rules": [
        { "when": { "class": "Editable Poly", "subobject": 2 },
          "menu": { "name": "Edges", "items": {
            "N": { "type": "macro", "category": "Editable Polygon Object", "name":
"EPoly_Connect" } } } }
      ]
    }
  }
}
```

Hover a command in the editor's catalogue to see its `table` and `id` - the fastest way to write actions by hand.

15. MAXScript commands

Type these in the MAXScript Listener (F11). They read state or change settings; none of them changes the scene except where said.

Command	Returns / does
<code>MarkingForge.version()</code>	The plugin version. "Unknown property" = the plugin is not loaded.
<code>MarkingForge.configSummary()</code>	The loaded file, the number of menus, errors and warnings.
<code>MarkingForge.configPath()</code>	The full path of menus.json.
<code>MarkingForge.reloadConfig()</code>	Reads menus.json again now.
<code>MarkingForge.menuModel <slot></code>	What dial <slot> holds, resolved: each item and whether it exists in this 3ds Max.
<code>MarkingForge.contextInfo()</code>	The class, kind, modifier and sub-object level the plugin sees now - the words to put into a condition.
<code>MarkingForge.windowInfo()</code>	The window under the cursor and the word for a window condition.
<code>MarkingForge.windowCatalog()</code>	Every window open now, with its word.
<code>MarkingForge.eventLog()</code>	The last gestures: which dial and variant opened, what ran or why it cancelled, how long it took.
<code>MarkingForge.clearEventLog()</code>	Empties the event log.
<code>MarkingForge.statsLine()</code>	Timing of the last dial opening.
<code>MarkingForge.mouseInfo()</code>	Mouse bindings and whether the mouse hook is installed.
<code>MarkingForge.usageInfo()</code> / <code>usageReset()</code>	The usage counters in one sentence / reset them.
<code>MarkingForge.experimentalInfo()</code>	Which experimental features are on.
<code>MarkingForge.experimentalGet "name"</code>	Whether one feature is on.
<code>MarkingForge.experimentalSet "name" true false</code>	Switches a feature and saves the file.
<code>MarkingForge.hotboxInfo()</code>	The hotbox's snapshot of the main menu.
<code>MarkingForge.lastChainDesc()</code> / <code>lastChainCode()</code>	The last chain you ran, as text / as MAXScript.
<code>MarkingForge.snapshotMenu <slot> @"C:\\dial.png" -1</code>	Saves a picture of dial <slot> as it would open now (the last number 0-7 draws that direction aimed, -1 at rest). Runs nothing.
<code>MarkingForge.sceneMenu...</code>	Chapter 13.

16. Tips and tricks

Learn by flicking, not by reading

Use the held dial for a few days, then start moving before it appears. Speed comes from never waiting for the drawing.

Put opposites opposite

Grow N / Shrink S, Hide S / Unhide N. The hand learns one movement and its reverse - the built-in dials follow this everywhere.

Keep NW as "back"

In every sub-object variant NW returns to object level. Reuse the convention in your own variants.

Eight is the limit, not the target

Put rarely used items in the list under the dial; directions are for what the hand should remember.

Short labels

Change label... with two words beats a cut-off 3ds Max name.

Colour the dial, not the item

A different dial colour per slot tells you which dial opened before you read a word. Colour single items only to warn - a red caption on a destructive command.

Specific variants first

Edit Poly at Vertex above Edit Poly above Editable Poly. When a variant never shows, a more general one above it is catching it first.

Let the context do the work

One key that changes with the sub-object level beats five keys. Look at the built-in Modelling dial before building your own.

Window variants

The same key can do one thing in the viewport and another over Slate or Track View - the built-in Materials, Animation and Scene dials show how.

Cross the rim for settings

Need the Chamfer caddy once? Move past the ring. Need it every time? Settings... on that item.

Chain several picks

Switch on *Multiple picks* first. Out past the ring and back to the dashed circle queues an item; release there runs the queue as one undo, the dot in the middle cancels. Save a chain you repeat as a single item with **Save last chain as an item...**

Let the dial suggest the next step

With pair learning on, MarkingForge notices what you run after what - Inset, then Extrude. Once a pair has repeated twice, the next command is the dial's outlined tile and a **tap** of the key runs it without aiming. Nothing on the dial moves; the memory lasts for the session, and **Diagnostics > Configuration status** shows how often the suggestions were taken.

Point instead of selecting

With *Menu under the cursor* the dial answers for the object you point at and the command runs on it - no click to select first. Pointing at one of several selected objects keeps the whole selection.

Type when you cannot find it

Switch on *Type-to-search*, hold any Alt or Shift+Alt dial and type a few letters: the dial becomes a search over every 3ds Max command. Enter runs the outlined best match.

Undo dial for big steps

Alt+7, aim NE or E: undo two or three steps in one move, by name.

Recent commands for repetition

Alt+8 repeats what you just did, newest first.

Selection sets under the hand

Shift+Alt+5 shows the scene's named selection sets as directions.

Hotbox for the rare command

Alt+2 shows all of 3ds Max's menus - including other plugins' - without hunting through the menu bar. Ctrl+Shift+middle mouse can open it too, if you bind it.

Scripts on a dial

Custom script... accepts any MAXScript. Snapshot the selection first (`local picked = selection as array`) when the code changes objects - the built-in dials do.

Find the words for a condition

`MarkingForge.contextInfo()` tells you the exact class and modifier names with your object selected.

Print the cheat sheet

Cheat sheet... in the editor makes a PDF of your saved dials to keep beside the keyboard for the first week; it tells you the page count when it is done.

Share a setup safely

Save preset... leaves out paths and usage; Load preset... strips scripts unless you allow them.

Back up your dials

Copy `menus.json`. It is a plain text file.

17. Problems and solutions

There is no MarkingForge menu after installing.

Restart 3ds Max - C++ plugins load only at start. Then type `MarkingForge.version()` in the Listener. If it says "Unknown property", check that `%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge\PackageContents.xml` exists and reinstall the Microsoft Visual C++ 2015-2022 x64 runtime.

3ds Max reports a duplicate plugin or class at start.

Two copies of `MarkingForge.gup` are installed. Run the installer: it lists every copy it finds. Keep only the one in `ApplicationPlugins`.

Holding the key does nothing.

Open **Customize > Hotkey Editor** and search the key: another command may own it, or the dial has no key. Run **MarkingForge > Set up standard shortcuts...** or set it in the editor and press Apply shortcuts.

The key opens a dial, but it is the wrong one / wrong contents.

A variant matched. `MarkingForge.eventLog()` names the variant that opened. With *Menu under the cursor* on, the object under the cursor decides - move the cursor to empty space or switch the feature off.

A direction has a red caption.

The command does not exist in this 3ds Max - usually a plugin that is not installed, or a macro from another machine. Replace the item in the editor.

A dial opens empty.

It uses a live source with nothing to show: the modifier stack needs exactly one selected object, selection sets need named sets in the scene, the undo/recent dials need their experimental switch on. The cheat sheet lists empty dials.

Save says the configuration changed after it was opened.

`menus.json` was changed on disk while the editor was open (another editor window, or a tick in **MarkingForge > Experimental features**). Choose **Save your edits as a separate copy**, press **Reload from file**, then carry your edits over.

Chamfer / Inset opens its settings when I do not want them (or the other way round).

Move past the rim to reverse it once; use **Settings...** on the item or `showItemSettings` to change the default.

An Edit Poly command does nothing.

Those commands act on the Edit Poly open in the Modify panel. Open the Modify panel with the modifier selected - the message in the Listener says so.

The dial stays on screen / a key feels stuck after switching programs.

Switching programs cancels the dial. If a key still seems held, press and release it once in 3ds Max.

Typing letters while the dial is open does something unexpected.

That is `gestureSearch`. Esc closes the search; switch the feature off if you do not use it.

A mouse binding does not work or breaks navigation.

Bindings need a modifier and a dial with its own menu. `MarkingForge.mouseInfo()` shows what is bound. Pick another combination if it collides with a 3ds Max navigation shortcut.

I edited menus.json and it stopped working.

Diagnostics > Configuration status names the line with the error; only that item is disabled. **Reload from file** in the editor shows the state after your fix.

The editor says "Some of the file could not be read".

A dial in `menus.json` is not a JSON object (a hand edit gone wrong). The editor lists which dials it left out; saving from the editor removes them. Fix the file by hand first if you want to keep them - the plugin skips the same dials and reports them in Configuration status.

An update failed half-way.

The installer backs up the installed version first and puts it back when copying fails, so MarkingForge keeps working as before; its message says so. Free some disk space or pause the antivirus and run **Update / Repair** again.

The editor does not open.

The message shows the Python error. Check that the `tools` folder is inside the plugin folder; reinstall with **Update / Repair**.

I want my shortcuts back as they were.

Every change leaves `plugcfg\MarkingForge\hotkeys\backup_*.hsx`. Load one in the Hotkey Editor, or use **Undo** next to Apply shortcuts right after a change.

I want the original dials back.

Editor > **Load default dials...**, tick *Replace the slots that already have a menu as well* to reset all.

Everything is broken and I want to start clean.

Close 3ds Max and rename `plugcfg\MarkingForge\menus.json`. The next start writes a fresh one with the built-in dials.

Still stuck? Open an issue at github.com/looki666/MarkingForge/issues or write to forgeplugins@gmail.com.

MarkingForge > Report a bug... puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

18. Files, privacy and safety

What	Where
Plugin	%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge (or %ProgramData%\... for all users)
Your dials	%LOCALAPPDATA%\Autodesk\3dsMax\2027 - 64bit\ENU\en-US\plugcfg\MarkingForge\menus.json
Hotkey set and backups	same folder, hotkeys\
Usage counters	same folder, usage.json
Startup log	same folder, menu_log.txt

- MarkingForge sends nothing over the network.
- It changes shortcuts only when you ask, and backs them up first.
- It writes nothing into scenes unless you store a scene menu.
- It never rearranges your dials by itself.
- Its keyboard hook exists only while a dial is open and touches no key outside 3ds Max.
- A **script** item runs with full rights, like any .ms or .mcr file - do not paste code you do not understand, and load presets from others with scripts removed (the default).

19. Support, bug reports and feature requests

ISSUES	github.com/looki666/MarkingForge/issues
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

- 1 In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
- 2 Open github.com/looki666/MarkingForge/issues and pick **Bug report** - or answer **Yes** in the dialog to write an e-mail to forgeplugins@gmail.com instead.
- 3 Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
- 4 Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.